

EXP NO: 04

ENCODER AND DECODER

DATE:
30/07/2026

ENCODER

Aim:

To construct and verify 9 to 4 encoder using IC 74147.

Apparatus Required:

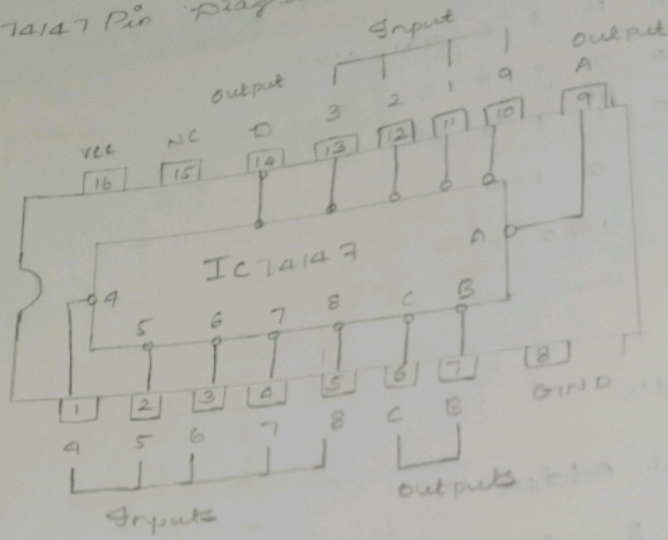
- 1) 9 to 4 Encoder - IC 74147.
- 2) connecting wires.
- 3) Digital trainer kit / breadboard.
- 4) +5V DC supply.

Theory:

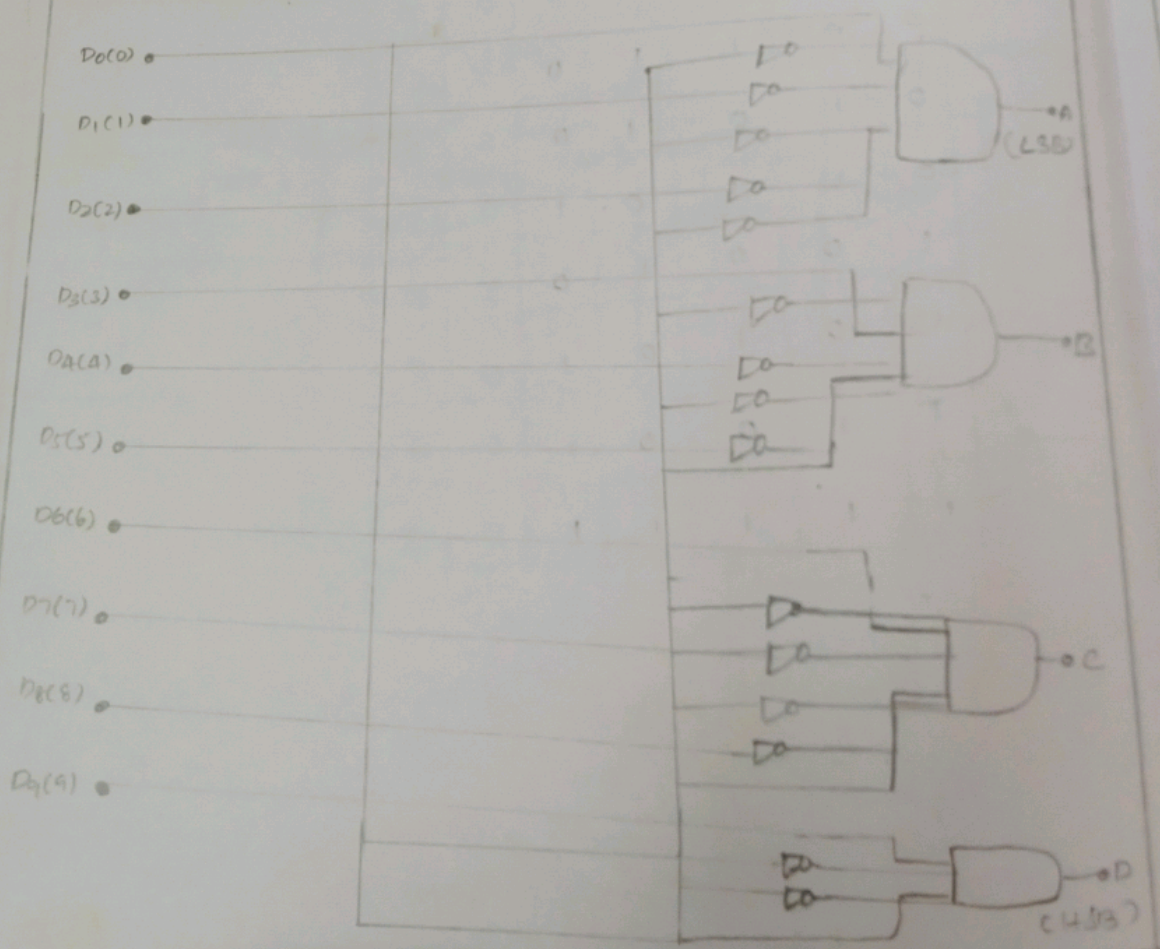
An encoder is a combinational circuit that converts information from a larger number of input line into a smaller number of output line.

A 9 to -4 Encoder has 9 input lines and 4 output lines. The inputs represent decimal digits 0 to 9 and corresponding BCD code at its output since the inputs and outputs of IC 74147 are active low, the selected input is connected to Low and the corresponding outputs are obtained in inverted form.

IC 74147 Pin Diagram



Circuit diagram



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ENCODER

Aim:

To convert
encoder

Apparatus

- 1) 9 to
- 2) Conn
- 3) Dig
- 4) +5

Theory:

An
circuit
large
small

A
and
dec
code
out
sel
+6

TRUTH TABLE / FUNCTION TABLE

INPUTS									OUTPUTS			
1	2	3	4	5	6	7	8	9	D	C	B	A
H	H	H	H	H	H	H	H	H	H	H	H	H
X	X	X	X	X	X	X	X	L	L	H	H	L
X	X	X	X	X	X	X	L	H	L	H	H	H
X	X	X	X	X	X	L	H	H	H	L	L	L
X	X	X	X	X	L	H	H	H	H	L	L	L
X	X	X	X	L	H	H	H	H	H	L	H	L
X	X	X	L	H	H	H	H	H	H	L	H	H
X	X	L	H	H	H	H	H	H	H	H	L	L
X	L	H	H	H	H	H	H	H	H	H	L	H
L	L	H	H	H	H	H	H	H	L	H	H	L

Procedure

- 1) Plan
- 2) Con
- 3) Co
- 4) C
- 5)
- 6)
- 7)

Procedure

- 1) Place the IC 74147 on the digital trainer kit
- 2) connected the IC to the required +5V supply and ground.
- 3) Connect the input switches to the input pins of the IC.
- 4) Give the Low to the required input corresponding to the decimal digit.
- 5) observe the four output lines.
- 6) Record the output for each input condition.
- 7) verify the obtained output with the function table.

Result:

Thus, the 9 to 4 encoder was constructed and verified using IC 74147.

DECODER

Aim:

To construct and verify 3×8 decoder using the IC 74138.

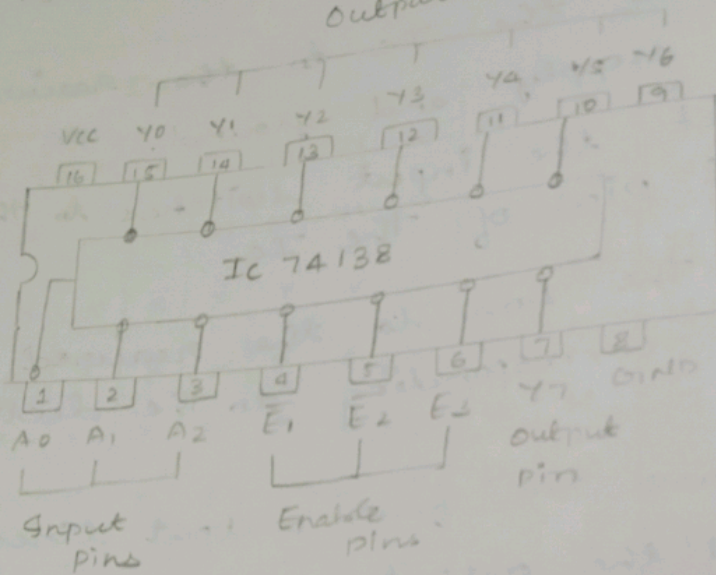
Apparatus Required:

- i) 3×8 decoder - IC 74138.
- ii) connecting wires.

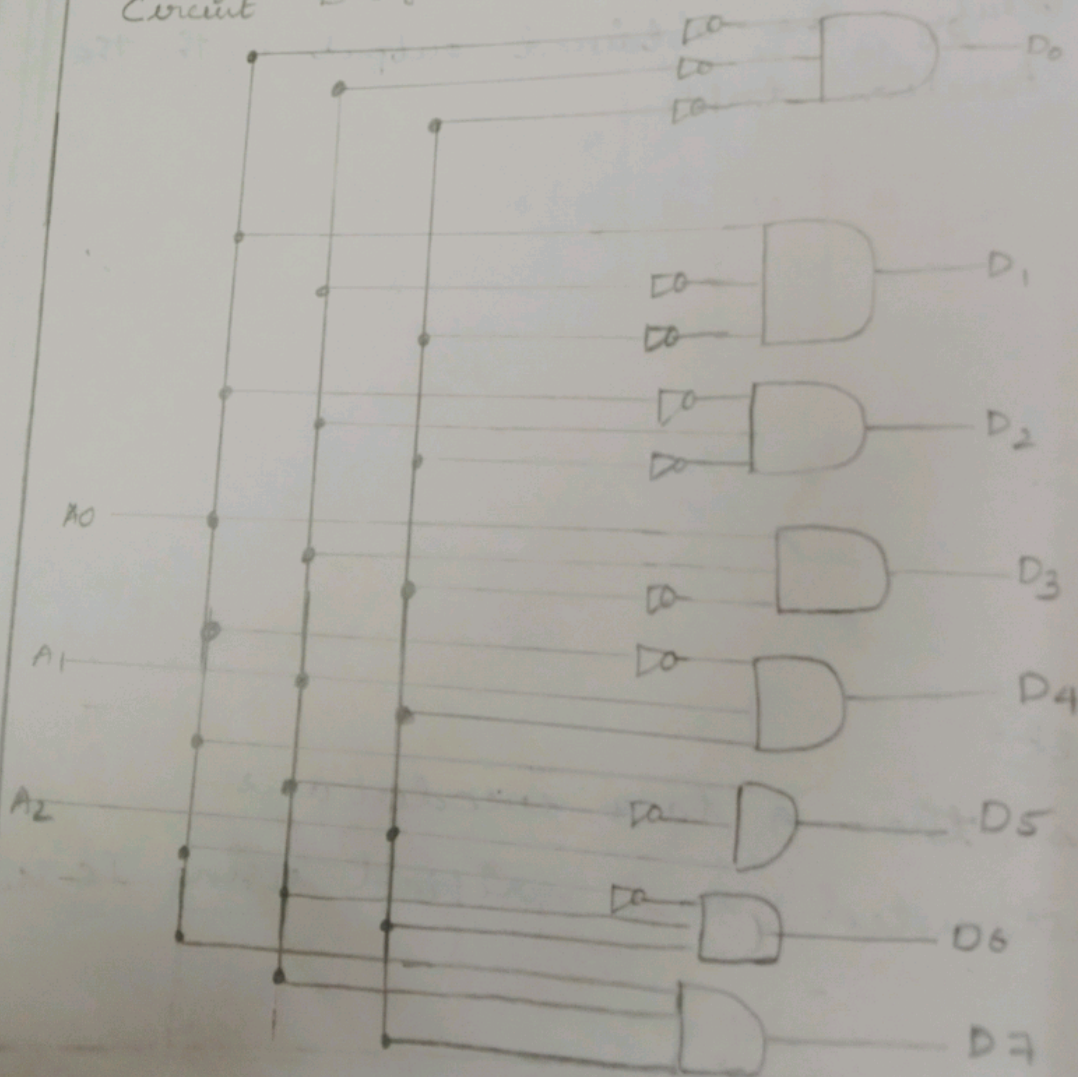
Theory:

Discrete quantities of information are present in digital systems by binary codes. A binary code of n bit can be represent n^{th} power of 2 information. A decoder is a combinational circuit that converts binary information from n input lines. These decoders are called n to m line decoders. The 3 inputs are decoded into 8 outputs each representing one of the minterms of 3 input variables. A particular application of this decoder is octal to binary conversion.

Pin diagram of IC 74138:
Output pins



Circuit Diagram



DECODE

AT

To con
using

Apparatu

- i) 3
- ii)

Theory

D
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A
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Truth Table / Function dr 1.10

[illegible]

Procedure:

- * The 3×8 decoder is mainly used for binary to octal conversion.
- * The circuit connection are given as per diagram
- * Inputs are given to 3 inverters.
- * Then each of the AND gates generates one of the minterms of the inputs.
- * Then the decimal ^{de}coded output is verified.

Result:

Thus the 3×8 decoder is constructed and verified using the digital IC 74138⁸.